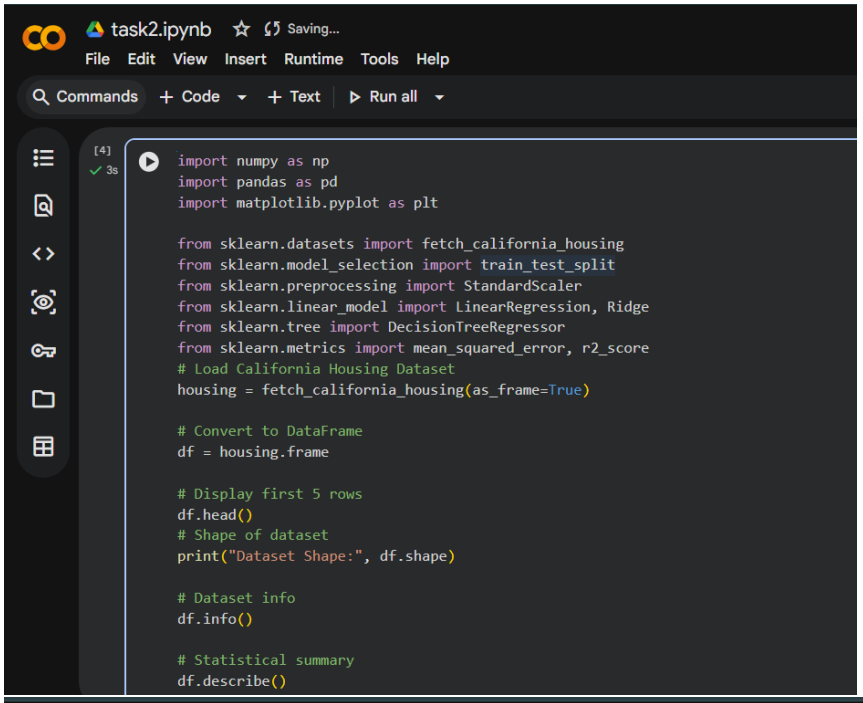
task2.ipynb
Saving...
File Edit View Insert Runtime Tools Help
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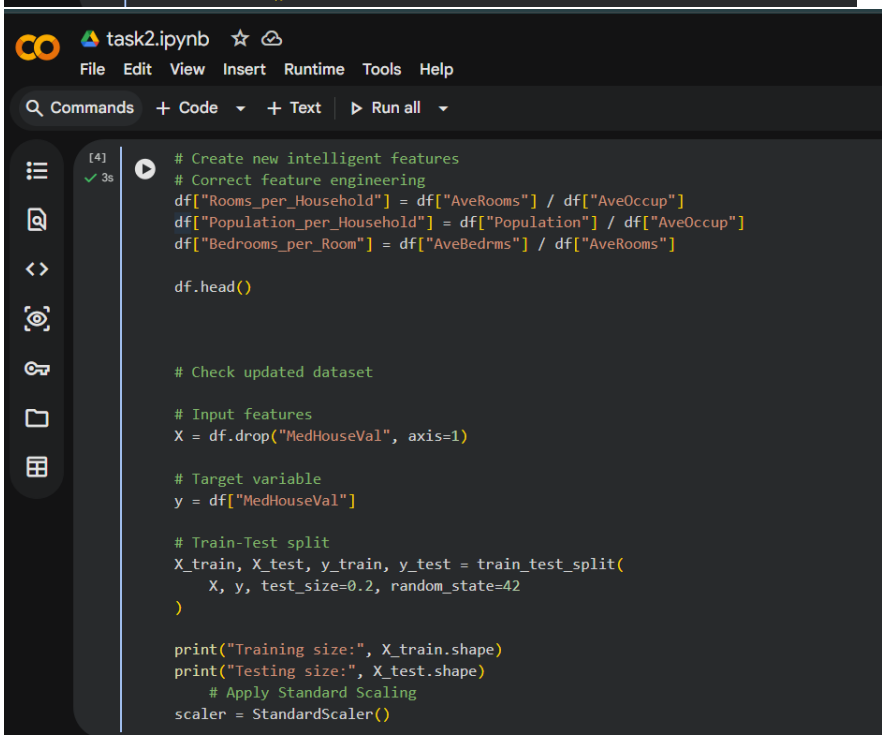
[4]
3s
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
# Load California Housing Dataset
housing = fetch_california_housing(as_frame=True)
# Convert to DataFrame
df = housing.frame
# Display first 5 rows
df.head()
# Shape of dataset
print("Dataset Shape:", df.shape)
# Dataset info
df.info()
# Statistical summary
df.describe()


task2.ipynb
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[4]
3s
Create new intelligent features
Correct feature engineering
df["Rooms_per_Household"] = df["AveRooms"] / df["AveOccup"]
df["Population_per_Household"] = df["Population"] / df["AveOccup"]
df["Bedrooms_per_Room"] = df["AveBedrms"] / df["AveRooms"]
df.head()
# Check updated dataset
# Input features
X = df.drop("MedHouseVal", axis=1)
# Target variable
y = df["MedHouseVal"]
# Train-Test split
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
print("Training size:", X_train.shape)
print("Testing size:", X_test.shape)
# Apply Standard Scaling
scaler = StandardScaler()

task2.ipynb

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[4] ✓ 3s

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)

# Predictions
lr_pred = lr_model.predict(X_test_scaled)

# Evaluation
lr_rmse = np.sqrt(mean_squared_error(y_test, lr_pred))
lr_r2 = r2_score(y_test, lr_pred)

print("Linear Regression RMSE:", lr_rmse)
print("Linear Regression R2 Score:", lr_r2)
# Ridge Regression
ridge_model = Ridge(alpha=1.0)
ridge_model.fit(X_train_scaled, y_train)

# Predictions
ridge_pred = ridge_model.predict(X_test_scaled)

# Evaluation
ridge_rmse = np.sqrt(mean_squared_error(y_test, ridge_pred))
ridge_r2 = r2_score(y_test, ridge_pred)

print("Ridge Regression RMSE:", ridge_rmse)
```

task2.ipynb

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[4] ✓ 3s

```
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices (Decision Tree)")
plt.show()
best_model = results.loc[results["RMSE"].idxmin()]
print("Best Performing Model:")
print(best_model)
plt.figure()
plt.bar(results["Model"], results["RMSE"])
plt.xlabel("Model")
plt.ylabel("RMSE")
plt.title("RMSE Comparison of Models")
plt.show()
plt.figure()
plt.bar(results["Model"], results["R2 Score"])
plt.xlabel("Model")
plt.ylabel("R² Score")
plt.title("R² Score Comparison of Models")
plt.show()
errors = y_test - dt_pred

plt.figure()
plt.hist(errors)
plt.xlabel("Prediction Error")
plt.ylabel("Frequency")
plt.title("Error Distribution (Decision Tree)")
plt.show()
plt.figure()
```

task2.ipynb

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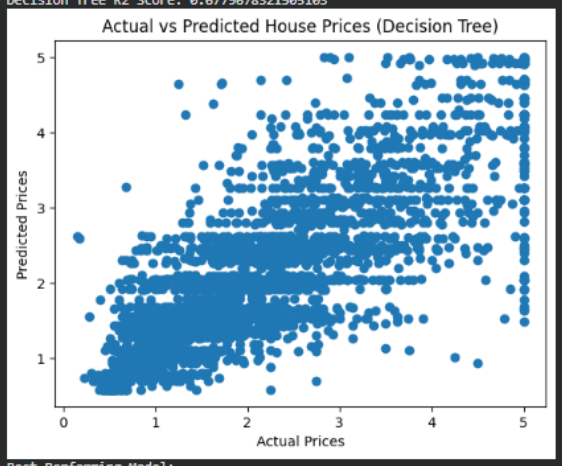
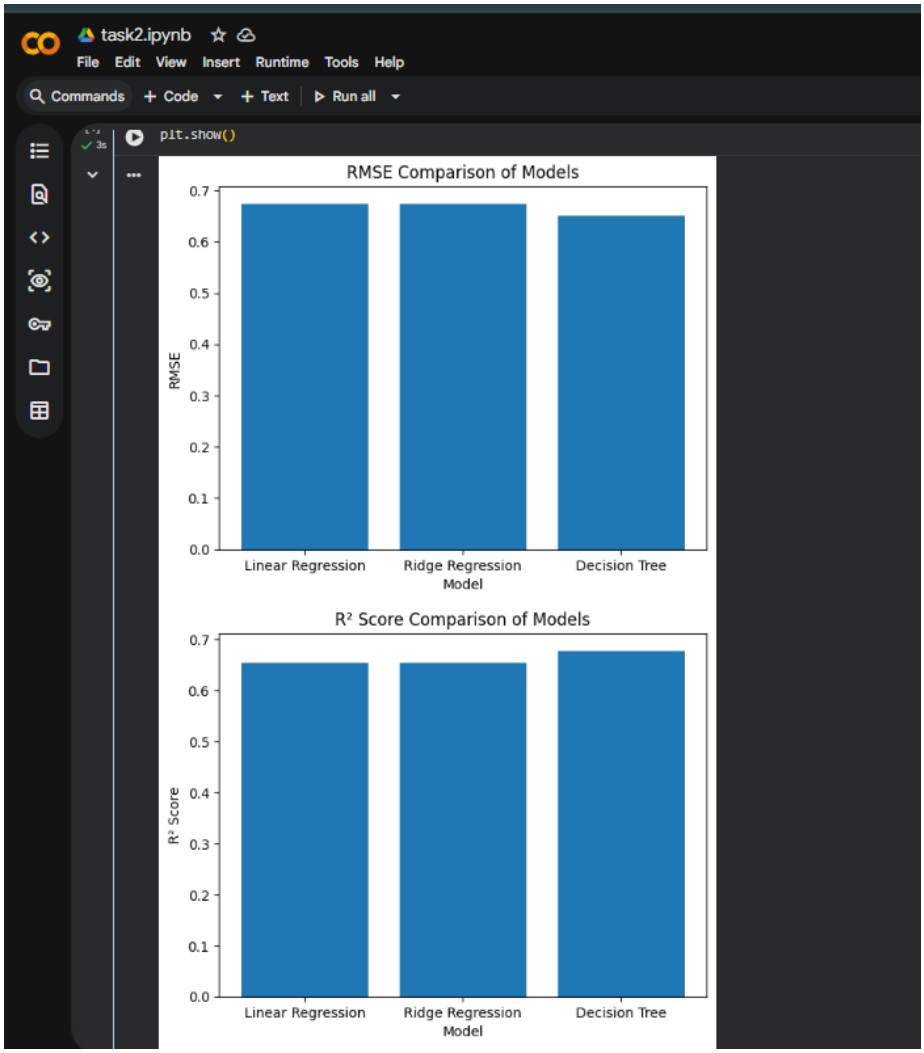
```
pit.show()
```

```
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   MedInc       20640 non-null  float64
 1   HouseAge     20640 non-null  float64
 2   AveRooms     20640 non-null  float64
 3   AveBedrms    20640 non-null  float64
 4   Population   20640 non-null  float64
 5   AveOccup     20640 non-null  float64
 6   Latitude     20640 non-null  float64
 7   Longitude    20640 non-null  float64
 8   MedHouseVal  20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
Training size: (16512, 11)
Testing size: (4128, 11)
Linear Regression RMSE: 0.6738172773395145
Linear Regression R2 Score: 0.6535205948827303
Ridge Regression RMSE: 0.6738201776284443
Ridge Regression R2 Score: 0.6535176121971285
Decision Tree RMSE: 0.649610532574982
Decision Tree R2 Score: 0.6779678321505103
```




```
Best Performing Model:
Model       Decision Tree
RMSE             0.649611
R2 Score         0.677968
Name: 2, dtype: object
```


task2.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
plt.show()
RMSE Comparison of Models
RMSE
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
Linear Regression
Ridge Regression
Decision Tree
Model
R² Score Comparison of Models
R² Score
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
Linear Regression
Ridge Regression
Decision Tree
Model

task2.ipynb

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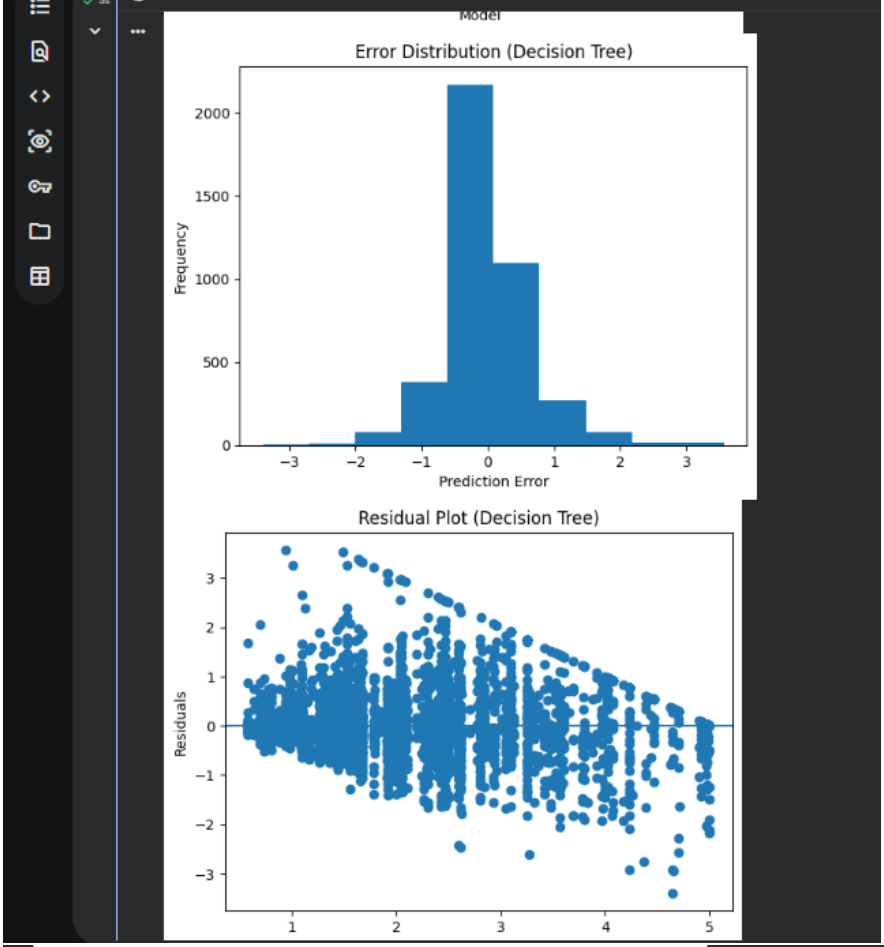
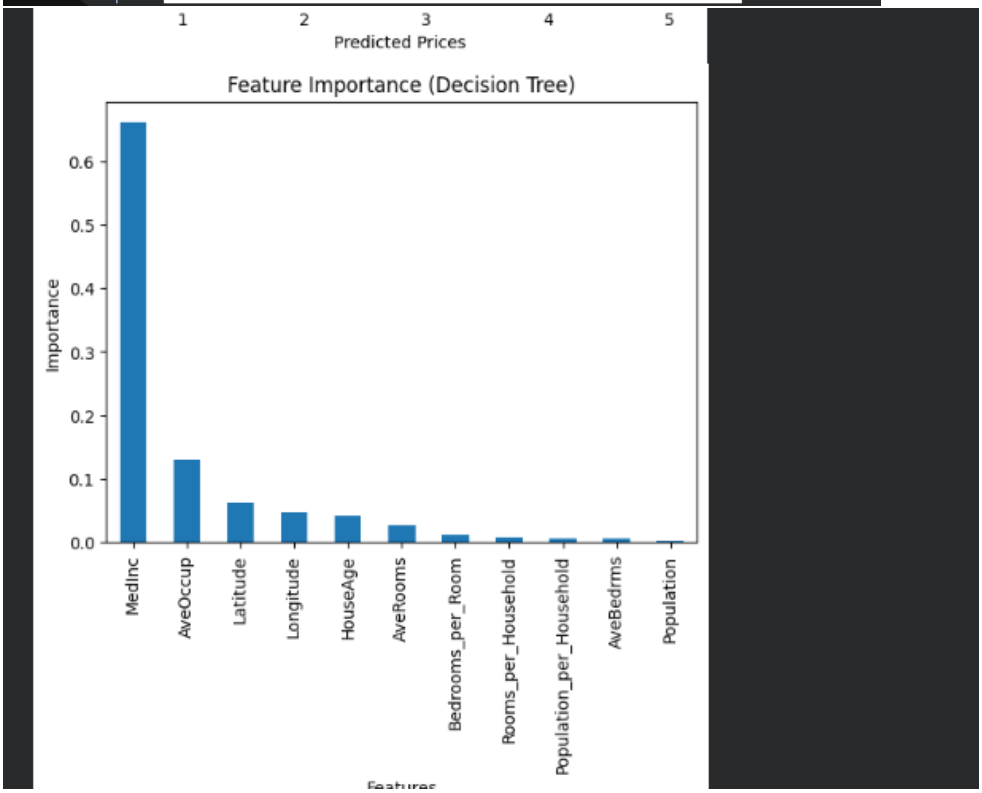
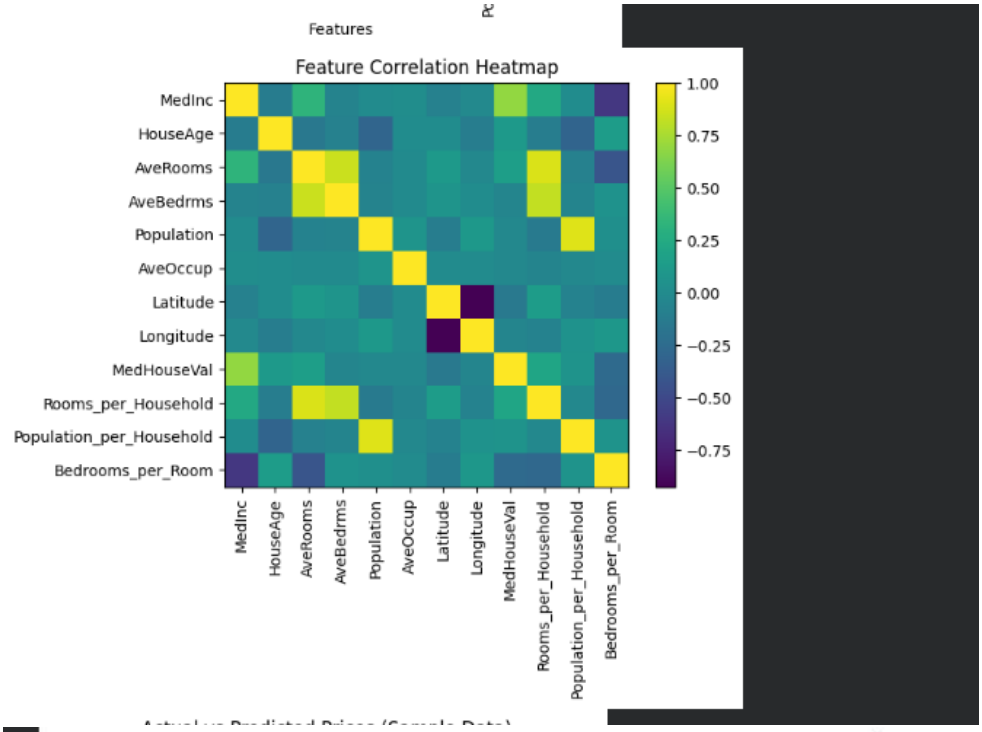
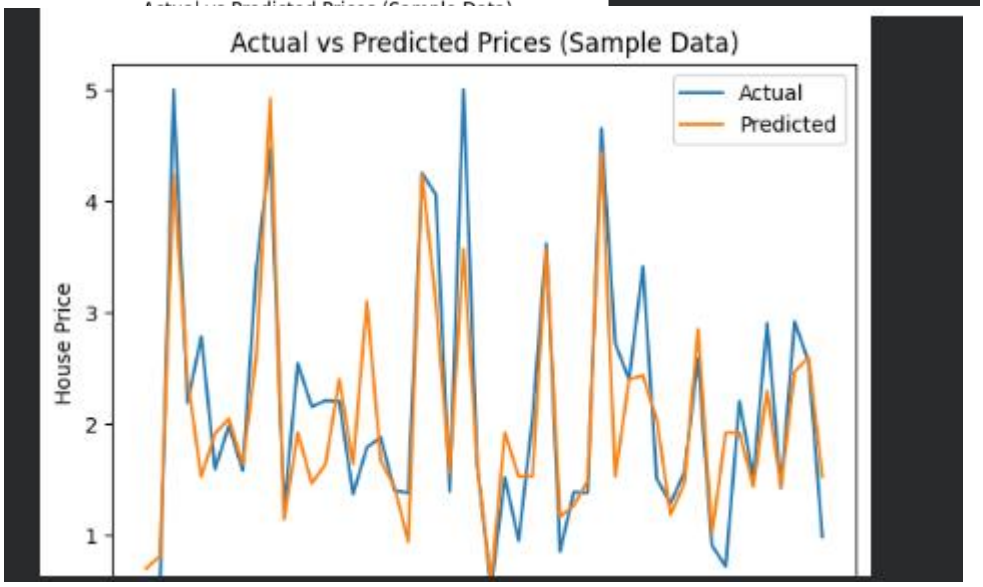
Commands + Code + Text Run all

```
plt.show()
```






Features
Feature Correlation Heatmap
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal
Rooms_per_Household
Population_per_Household
Bedrooms_per_Room
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Actual vs Predicted Prices (Sample Data)
Actual
Predicted
House Price
5
4
3
2
1